# MILITARY SPECIFICATION SHEET

## BATTERY, NON-RECHARGEABLE, LITHIUM THIONYL CHLORIDE BA-6704/U

This specification sheet is approved for use by U.S. Army Laboratory Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirement for acquiring the batteries described herein shall consist of this specification and the latest issue of MIL-B-49461(ER).

AMSC N/A

FSC 6135

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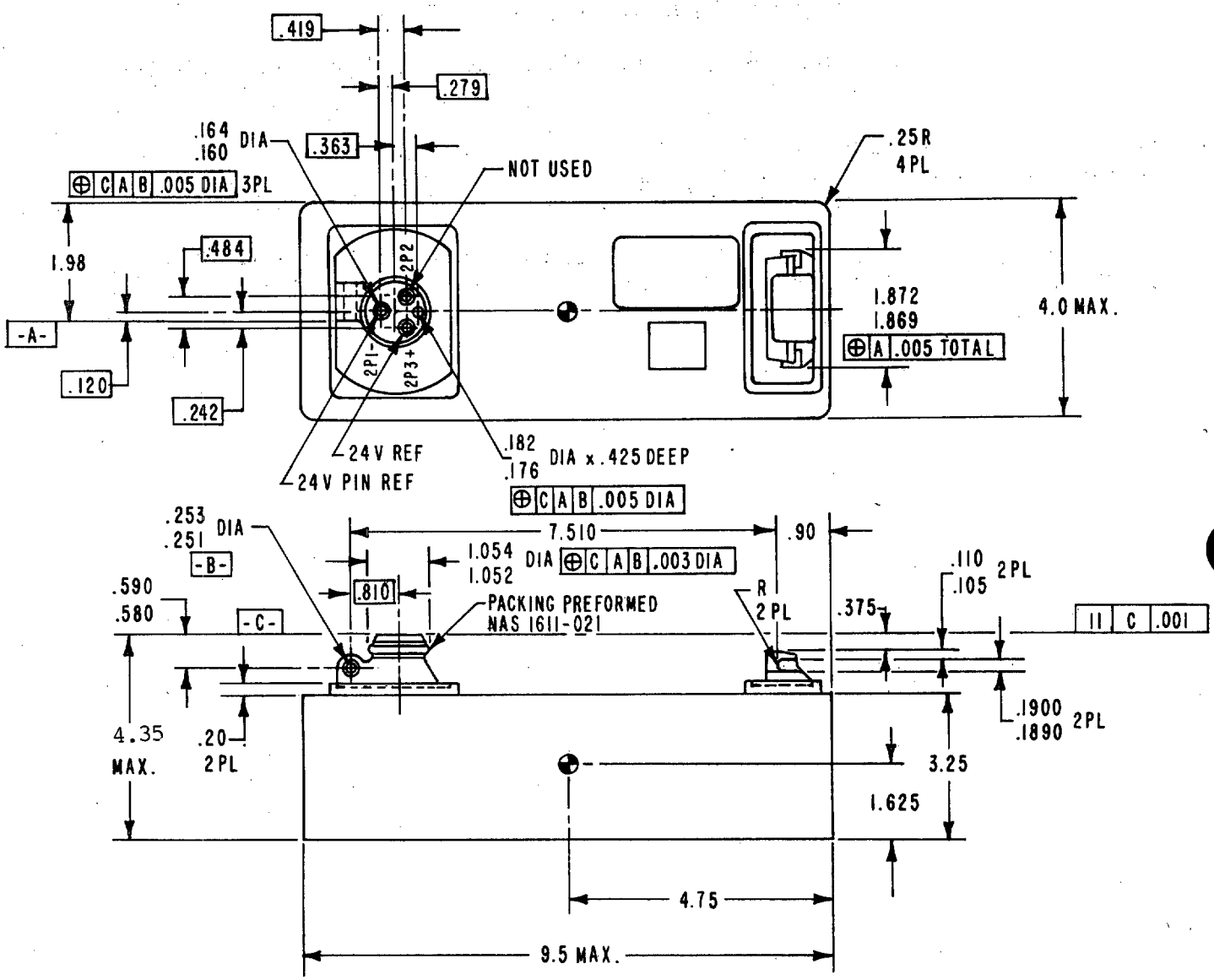



FIGURE 1. Battery Storage-Installation Control.



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NOTES

(1) Dimensions are in inches and unless otherwise specified decimal tolerances are ±.031 in.

(2) Identification label shall include complete label information required in the specification. It shall be located as shown on the drawing and the surface shall be designated as the top of the battery. If necessary, the adjacent side may contain continuation of label content. The identification label shall contain the following information:

Battery, Non-Rechargeable Lithium Thionyl Chloride, BA-6704/U
(Contract No.)
(Date Code) (Serial No.)
(Manufacturer)
(Location)

(3) Terminal markings. The positive (+) terminal marking may be painted red. The negative (-) terminal may be painted black. The two terminals shall be low resistance flat contacts of nickel per MIL-N-46025.

(4) A 20 ampere slow blow non-replaceable fuse shall be used.

(5) A thermal switch shall be physically located on each cell.

(6) Design and Construction: The battery shall consist of eight (8) series connected lithium thionyl chloride flat type cells sized for packaging in the battery envelope shown in Figure 1.

(7) The battery shall contain a blocking diode wired in series.

(8) The battery shall not weigh more than 7.5 lbs (3.4 kg.).

(9) ◒ denotes approximate center of gravity.



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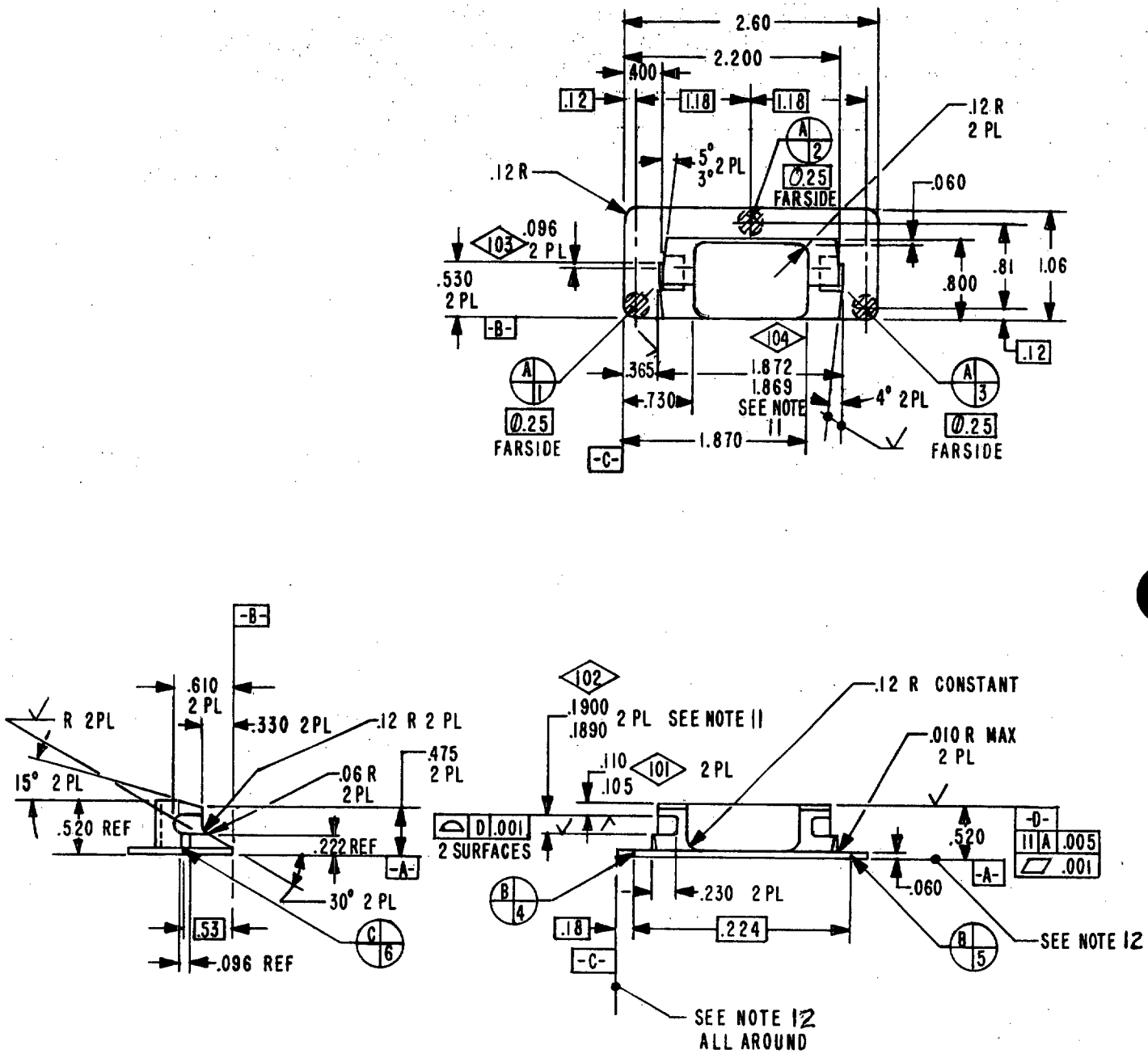



FIGURE 2. Bracket, Battery Pivot.



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NOTES

(1) Material: A356 A1 alloy casting per AMS 4218.

(2) Cast surfaces 125.

(3) All edge radii .015 maximum.

(4) Foundry welding permitted in accordance with MIS 23539, Type II.

(5) Casting shall be free of fins, sprues, risers, and gates.

(6) Features indicated by ✓ may be machined to conform to dimension and tolerance as shown.

(7) Alternate fabrication method permissible using bar, A1 alloy 6061-T6, QQ-A-22518, item 6.

(8) All fillet radii .031.

(9) Identify in accordance with MIL-STD-130.

(10) Finish in accordance with MIL-STD-186, code 203, black 27038 per FED-STD-595.

(11) Indicated dimensional limits apply after finishing, all others apply before finishing.

(12) Do not finish indicated surfaces.



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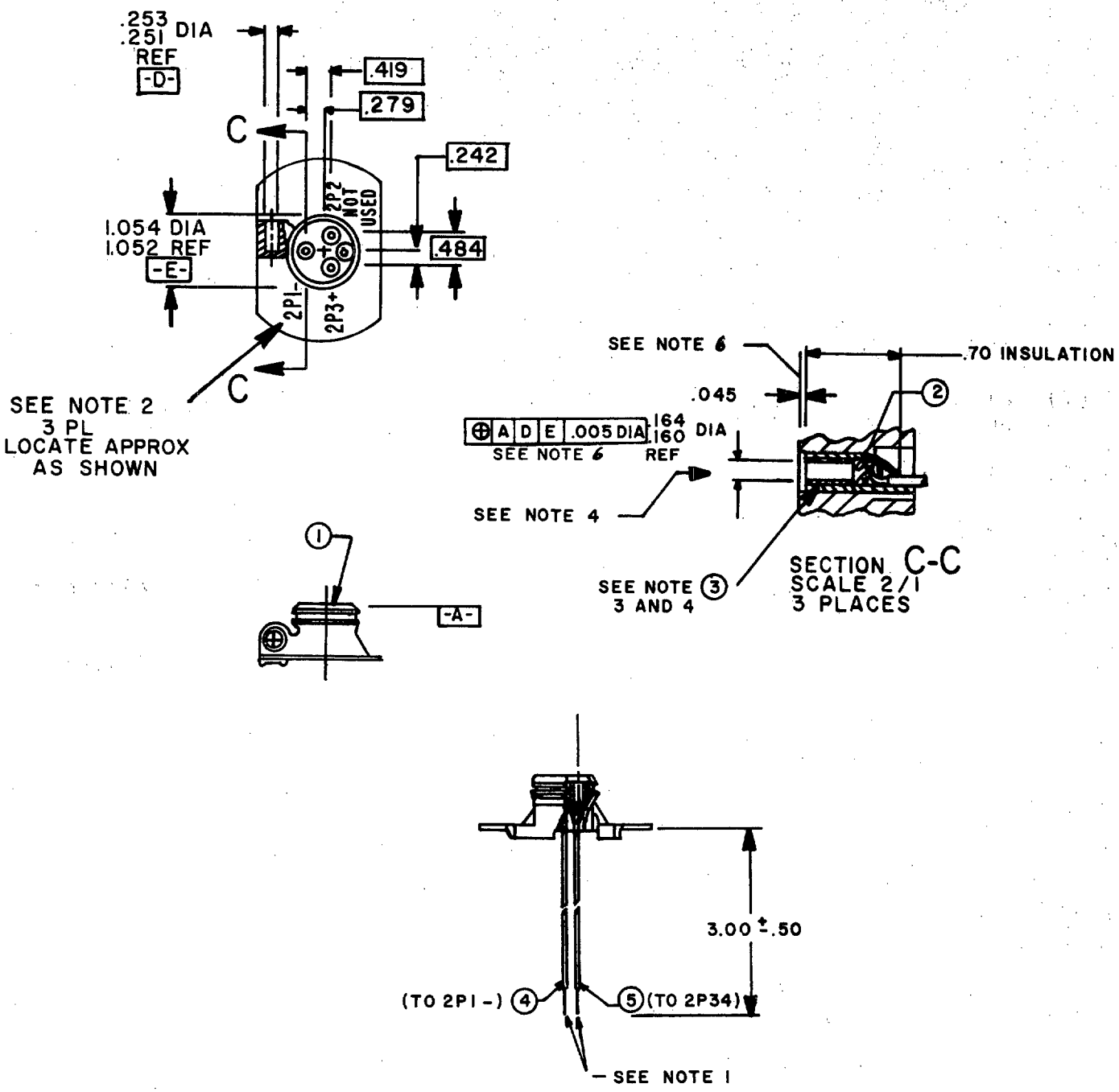



FIGURE 3. Receptacle Assembly.



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NOTES

(1) Solder wires per MIL-S-45743 using solder, Sn 60 or Sn 63 per QQ-S-571. Connect red wire per MIL-W-16878/4 to (+) terminal; connect black wire per MIL-W-16878/4 to (-) receptacle terminal.

(2) Mark characters using ink per finish code 806, MIL-STD-186, color lusterless white, .09 inch high, air dried.

(3) Install heat shrinkable insulation sleeving per MIL-I-23053/5 over terminal and wire as shown per MIS-30384.

(4) Bond sleeve covered terminal to receptacle shell per MIS-23608, type II, class A. Terminal shall not pull out when subjected to a 15 lbs force in indicated direction.

(5) Identify in accordance with MIL-STD-130.

(6) Surface shall be free of bond material.

Electrical Requirements for Battery

Battery Voltage:

No Load Voltage: The no load voltage shall not exceed 30 volts.

Operating Voltage: The nominal operating voltage shall be 20 volts.

Capacity Tests:

(1) Storage: Details of storage conditions for all capacity tests are specified in the basic specification.



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(2) Discharge: Batteries shall be pulse discharged at 20A for 30 milliseconds followed by 3.2A for 29 milliseconds, for a total of 3 minutes continuous cycling, then off 27 minutes. This 3 minutes on, 27 minutes off cycle shall be repeated continuously until zero volts is reached. However, time to 20V shall be used for calculation of capacity and/or service life. There shall be no venting, fire or explosives when the batteries are discharged.

| Capacity Tests | Battery Capacity in Amp-hours to 20.00 volts |
|---|---|
| I | 20 |
| L | 7.0 |
| H | 20 |
| HT | 18 |
| LT | 6.0 |
| T | 18 |
| D | 20 |
| IT | 18 |

Initial Voltage Delay: When the battery is subjected to any of the capacity tests, the time required at the beginning of discharge for the battery to reach a voltage of 20.00 volts after the specified load is applied shall not be more than 1.0 minutes.

Voltage Delay: A fixed resistor of 1.3 ohms shall be applied to the battery prior to the capacity discharge tests to measure voltage delay.

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